

Secure Shell (SSH)

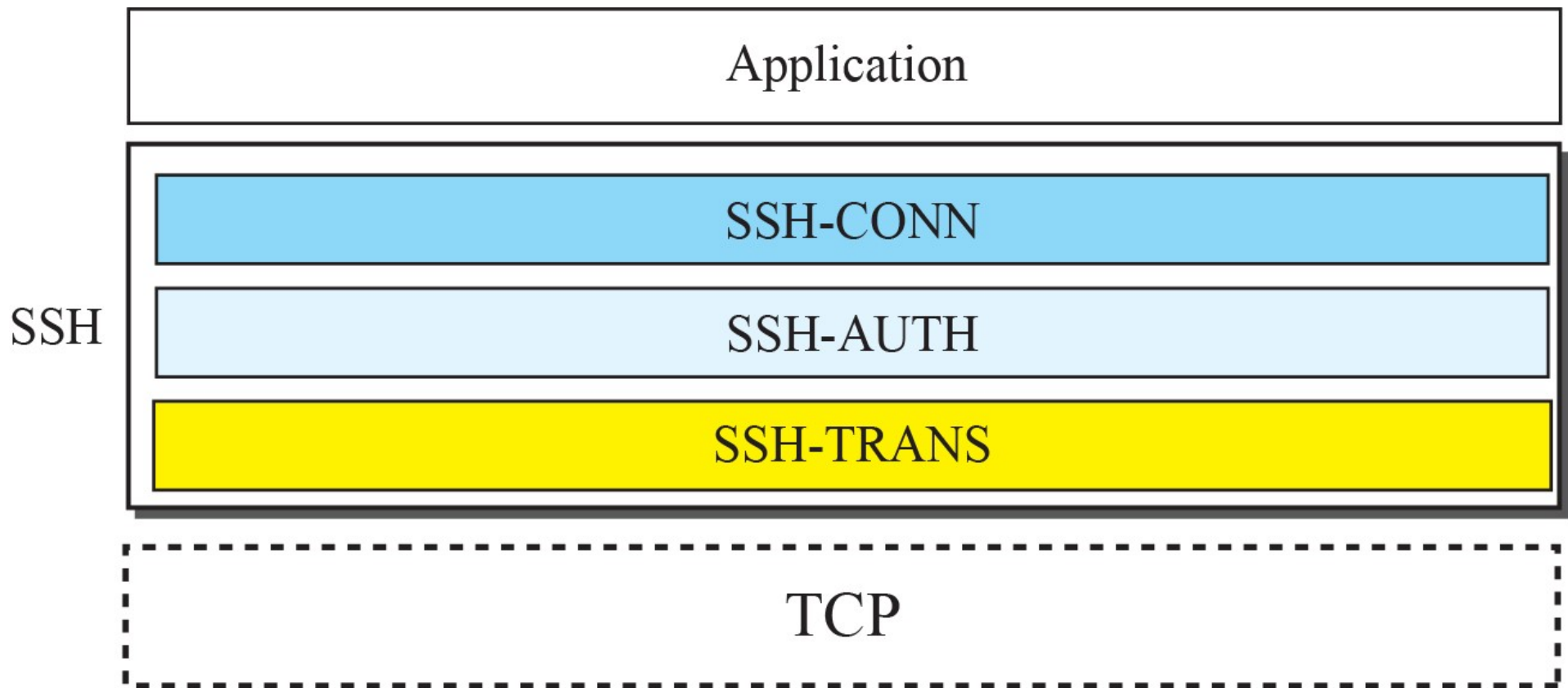


SECURE SHELL (SSH)

Although Secure Shell (SSH) is a secure application program that can be used today for several purposes such as remote logging and file transfer, it was originally designed to replace TELNET. There are two versions of SSH. The first version, SSH-1, is now deprecated because of security flaws in it. In this section, we discuss only SSH-2.

Components

SSH is an application-layer protocol with three components, as shown in Figure.



SSH Transport-Layer Protocol (SSH-TRANS)

- 1. Privacy or confidentiality of the message exchanged*
- 2. Data integrity, which means that it is guaranteed that the messages exchanged between the client and server are not changed by an intruder.*
- 3. Server authentication, which means that the client is now sure that the server is the one that it claims to be.*
- 4. Compression of the messages, which improves the efficiency of the system and makes attack more difficult.*

SSH Authentication Protocol (SSH-AUTH)

- 1. Authenticate the client for the server.*
- 2. defines a number of authentication tools similar to the ones used in SSL.*
- 3. Authentication starts with the client, which sends a request message to the server.*
 - The request includes the username, server name, the method of authentication, and the required data.*
- 4. The server responds with either a success message, which confirms that the client is authenticated, or a failed message.*

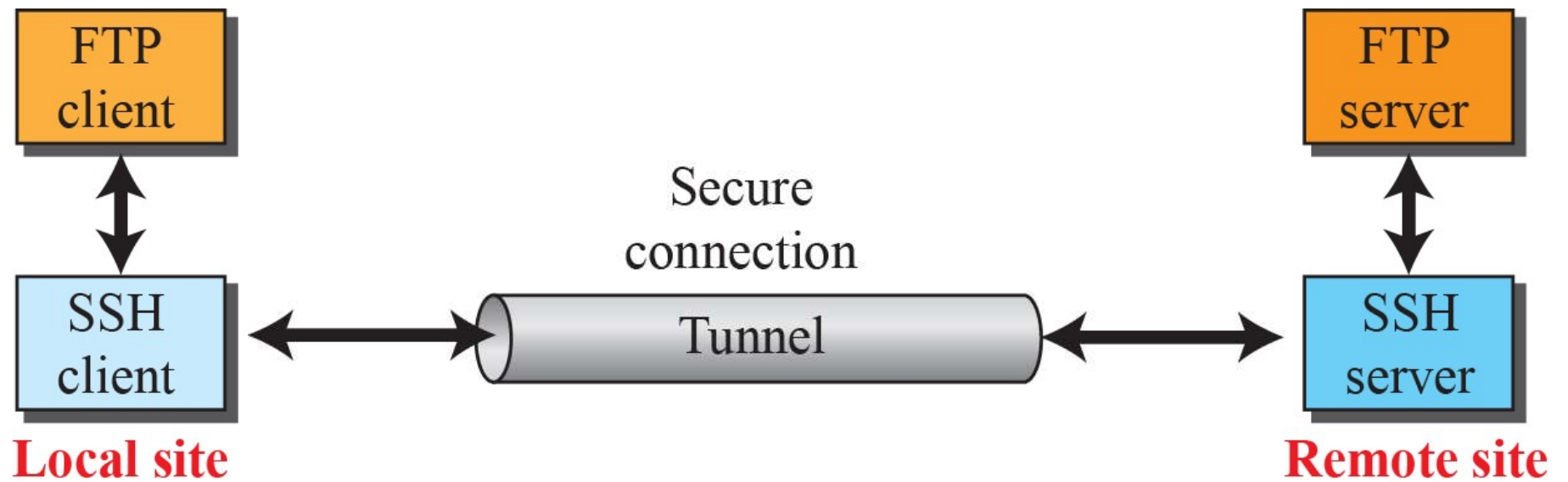
SSH Authentication Protocol (SSH-AUTH)

- 1. Multiplexing.*
- 2. defines a number of authentication tools similar to the ones used in SSL.*
- 3. Lets the client create multiple logical channels over it.*
- 4. Each channel can be used for a different purpose, such as remote logging, file transfer, and so on.*

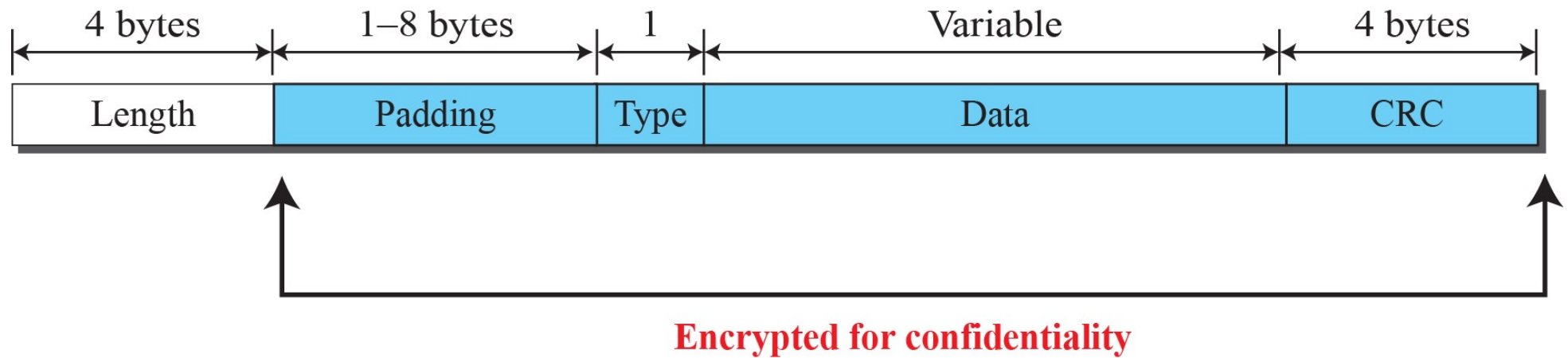
Applications

- *Although SSH is often thought of as a replacement for TELNET, SSH is, in fact, a general-purpose protocol that provides a secure connection between a client and server.*
- *PuTTY, by Simon Tatham, which is a client SSH program that can be used for remote logging.*
- *Application program is Tectia used on several platforms.*
 - *for file transfer is the Secure File Transfer Program (sftp)*
 - *Secure Copy (scp) - This application uses the same format as the UNIX copy command, cp, to copy files.*

SSH Tunneling - Port Forwarding



SSH Packet Format



Thank You...